

Subject: All of Statistics, Random Variables

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1 Random Variables

Random Variables

Random Variable. 随机变量 X 实际为一映射 (mapping) $X: \Omega \rightarrow \mathbb{R}$, 其给每一个 realization 赋以实值. 例如 $X(\omega)$ 表示抛 2 次硬币正面向上次数, 则若 $\omega = \{\text{正}, \text{反}\}$, 那么 $X(\omega) = 1$.

Note 1. 概率 P 是直接作用于事件 E (即可能发生情况的某个集合) 而非随机变量上的, 而满足 $X(\omega) = x, X(\omega) \in A$ 的 ω 均可以形成相应集合, 即 $X^{-1}(x) = \{\omega \in \Omega: X(\omega) = x\}, X^{-1}(A) = \{\omega \in \Omega: X(\omega) \in A\}$, 因此也可以对随机变量应用概率 P :

$$P(X \in A) = P(\{\omega \in \Omega: X(\omega) \in A\}), \quad P(X = x) = P(\{\omega \in \Omega: X(\omega) = x\}).$$

Distribution Function. ① Cumulative distribution function (CDF): 累积分布函数 $f_X: \mathbb{R} \rightarrow [0, 1]: F_X(x) = P(X \leq x)$. ② $F: \mathbb{R} \rightarrow [0, 1]$ 为 CDF $\Leftrightarrow$ (1) F 非单调递减 ($x_1 \leq x_2 \implies F(x_1) \leq F(x_2)$) (2) $\lim_{x \rightarrow -\infty} F(x) = 0, \lim_{x \rightarrow \infty} F(x) = 1$ (3) F 右连续: $F(x) = F(x^+), F(x^+) = \lim_{y \rightarrow x, y > x} F(y)$.

Probability Function. ① Discrete: X 的取值可数即离散时定义概率函数 (probability function) $f_X(x) = P(X = x)$, 满足 $F_X(x) = P(X \leq x) = \sum_{x_i \leq x} f_X(x_i)$. ② Continuous: X 连续时定义概率密度函数 (probability density function, PDF) f_X 满足 $f_X(x) \geq 0, \int_{-\infty}^{\infty} f_X(x) dx = 1, F_X(x) = \int_{-\infty}^x f_X(t) dt, f_X(x) = F'_X(x)$.

Note 2. ① 只有当随机变量 X 为离散时概率函数 f_X 才理解为概率 P , 当 X 连续时 $\forall x, P(X = x) = 0$, f 只满足在区间上积分为 1, 例如 $f(x) = 5, x \in [0, 1/5]$. ② 随机变量分布相同并不代表两个随机变量完全相同, 例如 $P(X = 1) = P(X = -1) = 1/2, Y = -X$.

Independent Random Variables. 定义量随机变量 X, Y 为相互独立的若 $\forall A, B, P(X \in A, Y \in B) = P(X \in A) \cdot P(Y \in B)$.

Independent and Identical Distribution. 对于相互独立的随机变量 $X_1, \dots, X_n$ 若它们服从相同的分布 F , 则称它们独立同分布(i.i.d.): $X_1, \dots, X_n \sim F$.

Important Distribution – Discrete

记随机变量 X 服从分布 F 为 $X \sim F$, 一些常用和重要的离散分布如下:

1. Point Mass Distribution $x \sim \delta_a: P(X = a) = 1, F(x) = \begin{cases} 0, & x < a; \\ 1, & x \geq a. \end{cases}, f(x) = \begin{cases} 1, & x = a; \\ 0, & x \neq a. \end{cases}$
2. Discrete Uniform Distribution: 离散均匀分布, $f(x) = \begin{cases} 1/n, & \text{for } x = 1, \dots, n; \\ 0, & \text{otherwise.} \end{cases}, n \in \mathbb{N}_+$
3. Bernoulli Distribution $X \sim \text{Bernoulli}(p)$: 伯努利分布, $P(X = 1) = p, P(X = 0) = 1 - p, p \in [0, 1], f(x) = p^x(1 - p)^{1-x}, x \in \{0, 1\}$.
4. Binomial Distribution $X \sim \text{Binomial}(n, p)$: 二项式分布,

$$f(x) = \begin{cases} \binom{n}{x} p^x (1 - p)^{n-x}, & \text{if } x = 0, 1, \dots, n, \\ 0, & \text{otherwise.} \end{cases}$$

满足若 $X_1 \sim \text{Binomial}(n_1, p), X_2 \sim \text{Binomial}(n_2, p) \implies X_1 + X_2 \sim \text{Binomial}(n_1 + n_2, p)$

5. Geometric Distribution $X \sim \text{Geom}(p)$: 几何分布, $P(X = k) = p(1 - p)^{k-1} = f(k), k \in \mathbb{N}_+$.
6. Poisson Distribution $X \sim \text{Poisson}(\lambda)$: 泊松分布, $f(x) = e^{-\lambda} \frac{\lambda^x}{x!}, x \in \mathbb{N}_0$. 满足若 $X_1 \sim \text{Poisson}(\lambda_1), X_2 \sim \text{Poisson}(\lambda_2)$ 则 $X_1 + X_2 \sim \text{Poisson}(\lambda_1 + \lambda_2)$.

Important Distribution – Continuous

一些常用和重要的连续分布如下:

1. Uniform Distribution $X \sim U[a, b]$: 均匀分布:

$$f(x) = \begin{cases} \frac{1}{b-a}, & x \in [a, b] \\ 0, & \text{otherwise} \end{cases}, \quad F(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & x \in [a, b] \\ 1 & x > b. \end{cases}$$

2. Normal (Gaussian) Distribution $X \sim \mathcal{N}(\mu, \sigma^2)$: $f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), x \in \mathbb{R}$.

3. Standard Normal Distribution: $X \sim \mathcal{N}(0, 1)$, 并特定地记此时 X 为记号 Z , 记 PDF f 为 $\phi(z)$, CDF F 为 $\Phi(z)$. 满足(1) $X \sim \mathcal{N}(\mu, \sigma^2) \implies Z = (X - \mu)/\sigma \sim \mathcal{N}(0, 1)$ (2) $Z \sim \mathcal{N}(0, 1) \implies X = \mu + \sigma Z \sim \mathcal{N}(\mu, \sigma^2)$ (3) $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$ 互相独立 $\implies \sum_{i=1}^n X_i \sim \mathcal{N}(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2)$.

4. Exponential Distribution $X \sim \text{Exp}(\beta)$: $f(x) = \frac{1}{\beta} e^{-x/\beta}, x > 0, \beta > 0$.

5. Gamma Distribution $X \sim \text{Gamma}(\alpha, \beta)$: $f(x) = \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, x > 0, \alpha, \beta > 0$, 其中 $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx, \alpha > 0$ 为 Gamma 函数. (1) $\text{Gamma}(1, \beta) = \text{Exp}(\beta)$ (2) 若 $X_i \sim \text{Gamma}(\alpha_i, \beta)$ 互相独立, 则 $\sum_{i=1}^n X_i \sim \text{Gamma}(\sum_{i=1}^n \alpha_i, \beta)$.

6. Beta Distribution $X \sim \text{Beta}(\alpha, \beta)$: $f(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}, 0 < x < 1, \alpha, \beta > 0$.

7. t Distribution $X \sim t_\nu$: $f(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})} \frac{1}{(1+\frac{x^2}{\nu})^{(\nu+1)/2}} \cdot (\nu = \infty \text{ 时对应 Normal Distribution.})$

8. Cauchy Distribution: $\nu = 1, f(x) = \frac{1}{\pi(1+x^2)}$.

9. χ^2 Distribution $X \sim \chi_p^2$: $f(x) = \frac{1}{\Gamma(p/2)2^{p/2}} x^{(p/2)-1} e^{-x/2}, x > 0$. 满足若 $Z_i \sim \mathcal{N}(0, 1)$ 互相独立则 $\sum_{i=1}^p Z_i^2 \sim \chi_p^2$.

Bivariate, Marginal and Conditional Distribution

Bivariate Distribution. ① 对于一组离散随机变量 X, Y , 其联合概率函数 (joint mass function) 定义为 $f_{X,Y}(x) = P(X = x, Y = y)$. ② 对于一组连续随机变量 X, Y , 其联合概率密度函数 (joint PDF) $f(x, y)$ 满足: (1) $\forall x, y, f(x, y) \geq 0$ (2) $\int_{-\infty}^\infty \int_{-\infty}^\infty f(x, y) dx dy = 1$ (3) $P((X, Y) \in A) = \int \int_A f(x, y) dx dy, A \subseteq \mathbb{R} \times \mathbb{R}$.

Marginal Distributions. ① 对于一组离散随机变量 X, Y 及已知的 $f_{X,Y}$, 定义 X (Y 同理) 的边缘概率函数 (marginal mass function) 为 $f_X(x) = P(X = x) = \sum_y P(X = x, Y = y) = \sum_y f(x, y)$. ② 对于一组连续随机变量 X, Y 及已知的 $f_{X,Y}$, 定义 X (Y 同理) 的边缘概率密度函数为 $f_X(x) = \int f(x, y) dy$.

Note 3. ① X 与 Y 相互独立 $\Leftrightarrow \forall x, y, f_{X,Y}(x, y) = f_X(x)f_Y(y)$. ② (X, Y) 的定义区间为一矩形(可

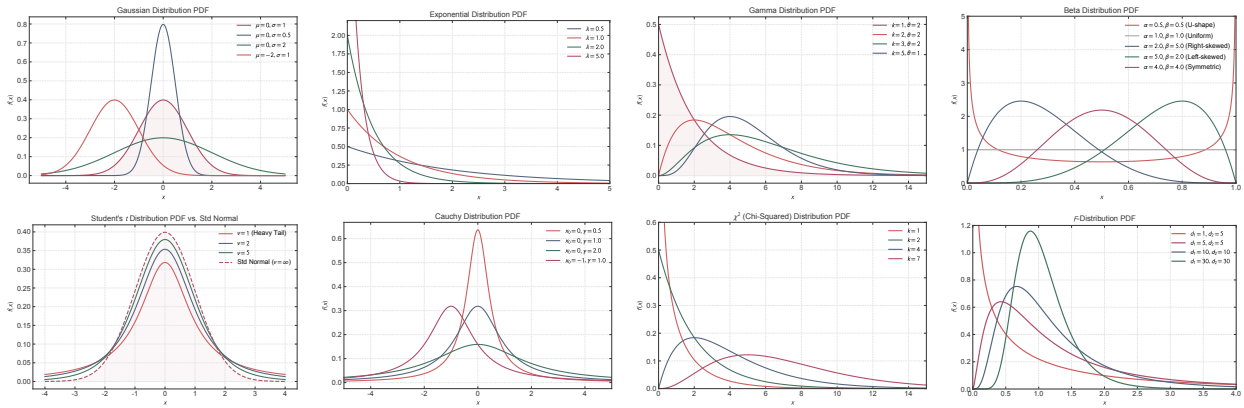
为 ∞), $f(x, y)$ 可写为 $g(x) \cdot h(y) \implies X, Y$ 相互独立.

Conditional Distribution. ① 对于一组离散随机变量 X, Y , $f_{X|Y}(x|y) = P(X = x|Y = y) = \frac{P(X=x, Y=y)}{P(Y=y)}$. ② 对于一组连续随机变量 X, Y , $f_{X|Y}(x|y) = P(X = x|Y = y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$, $f_Y(y) > 0$. 因此 $f_{X,Y}(x, y) = f_{X|Y}(x|y)f_Y(y) = f_{Y|X}(y|x)f_X(x)$.

Transformations of Several Random Variables

Transformation of Random Variables. 称 $Y = r(X)$ 为随机变量 X 的一个变换 (transformation). ① Discrete: 在离散情形下 $f_Y(y) = P(Y = y) = P(r(X) = y) = P(\{x : r(x) = y\}) = \sum_{x_i: r(x_i)=y} f_X(x_i)$. ② Continuous: 在连续情形下首先计算 CDF: $F_Y(y) = P(Y \leq y) = P(r(X) \leq y) = P(X \in A_y) = \int_{A_y} f_X(x)dx$, 其中 $A_y = \{x : r(x) \leq y\}$, 再计算 PDF: $f_Y(y) = F'_Y(y)$.

Transformations of Several Random Variables. 对于融合了多个随机变量的变换如 $Z = r(X, Y)$, 首先计算 CDF: $F_Z(z) = P(Z \leq z) = P(r(X, Y) \leq z) = P(\{(x, y); r(x, y) \leq z\}) = \int \int_{A_z} f_{X,Y}(x, y) dx dy$. 再计算 PDF $f_Z(z) = F'_Z(z)$.



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